K Praveen Reddy

LinkedIn: linkedin.com/in/praveen-reddy-- Email: praveenreddykalavapalli@gmail.com
GitHub: https://github.com/praveenreddy-kalavapalli Mobile: +91-9394837363

Skills

- **Languages**: C, C++, Python, Java,
- **Frontend:** HTML and CSS, Bootstrap
- **Tools/Platforms**: MySQL, Power BI, Excel
- **DevOps**: GitHub Actions(CI/CD), Linux/Ubuntu
- **CS Concepts**: DBMS, OS, CN, OOP, SQL
- **Soft Skills**: Problem-Solving Skills, Team Player, Project Management, Decision Making, Quick Learner, Adaptability

Projects

Dynamic Contact Tracing App using Graphs — JavaFX, AES Encryption **August 2025**

- Developed a JavaFX-based animated application to trace malware spread across IoT nodes using graph data structures.
- Implemented Breadth-First Search (BFS) to compute the shortest breach path between IoT devices in real-time.
- Built a 2D animated visualization engine that displays malware propagation using JavaFX shapes, transitions, and dynamic path rendering.
- Integrated AES encryption/decryption to securely log breach paths and protect simulation data.
- Designed and structured modular components (Graph, IoT Simulator, Encryptor) enabling scalable and maintainable system architecture.

Quantum Neural Networks (QNN) — Python **April 2025**

- Implemented a quantum neural network (QNN) simulation in Python.
- Used hybrid quantum-classical architecture to encode classical data into quantum representations and train a variational quantum circuit.
- Carried out parameter optimization using classical optimizers to tune the quantum circuit, demonstrating a basic quantum learning model.
- Validated performance on a sample dataset (e.g., segmented_customers_manual.csv) included in the repo.
- Explored quantum machine-learning principles like superposition and parameterized gates in a simulated environment (since this is classical simulation of quantum models).

Training

Summar Training | Lovely Professional University **Jun 2025-Jul 2025**

- Developed a Stack-based Browser History Manager demonstrating strong command over C++, OOP concepts, and Data Structures.
- Implemented core data structures (Stack, Queue, Linked List, Trees, Graphs) and algorithms (DFS, BFS, Sorting, Searching).
- Hands-on experience with STL (vectors, stacks, queues, maps) for writing optimized and clean C++ code.
- Built multiple mini projects applying inheritance, polymorphism, dynamic memory handling, file I/O, and pointer manipulation.

Certificates/ Achievements

- The Bits and Bytes of Computer Networking – Google **Sep 7, 2024**
- Computational Theory: Language Principle & Finite Automata Theory – Infosys Springboard **August 19, 2025**
- C++ Programming: OOPs and DSA – CSE Pathshala **August 3, 2025**
- Modern Android Development Skills by Google
- Fundamentals of Network Communication- Coursera **Sep 27, 2024**

Education

- **Lovely Professional University** **Punjab, India**
  Bachelor of Technology – CSE (Ai& ML) CGPA: 5.64 **Aug** 2023-Present
- **Narayana Junior College** **Andhra Pradesh**
  Intermediate Andhra Pradesh Board (89.8%) **May** 2021-2023
- **Bhashyam High School** **Guntur, India**
  Matriculation; Percentage: 100% **May** 2022-2021